SEFU HMIS — National Interoperability, Client Registry & REST API Specification

Document ID: SEFU-DHA-INT-API-001

Target Regulation: Kenya Digital Health Agency (DHA) HMIS Certification Framework (Form HMIS 4, Section G) & Kenya Data Protection Act 2019

System Name: SEFU Health Management Information System (HMIS)

Version: 1.0

Effective Date: November 2026

Classification: Confidential — Technical Certification Dossier

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1. Executive Overview & Interoperability Objectives

SEFU HMIS incorporates an authenticated, standardized JSON REST API gateway designed to guarantee bidirectional interoperability with Kenya's National Health Information Exchange (HIE).

This interface satisfies all regulatory criteria set forth under:

1. **Kenya Digital Health Agency (DHA) Certification Framework (Form HMIS 4):** Section G — "Interoperability, REST APIs, and System Integration".
2. **Kenya National E-Health Interoperability Framework:** Fast Healthcare Interoperability Resources (FHIR) demographic and clinical alignment.
3. **Kenya Client Registry (CR) Standard:** Unique Patient Identifier (UPI / Afya ID) and National ID lookup and ingestion.
4. **Kenya Master Health Facility List (KMHFL):** Canonical MFL facility identification.
5. **Kenya Health Worker Registry:** Practitioner licensing attribution (KMPDC / NCK).
6. **Social Health Authority (SHA) Claims Gateway:** Pre-authorization queries and itemized financial claim exports.

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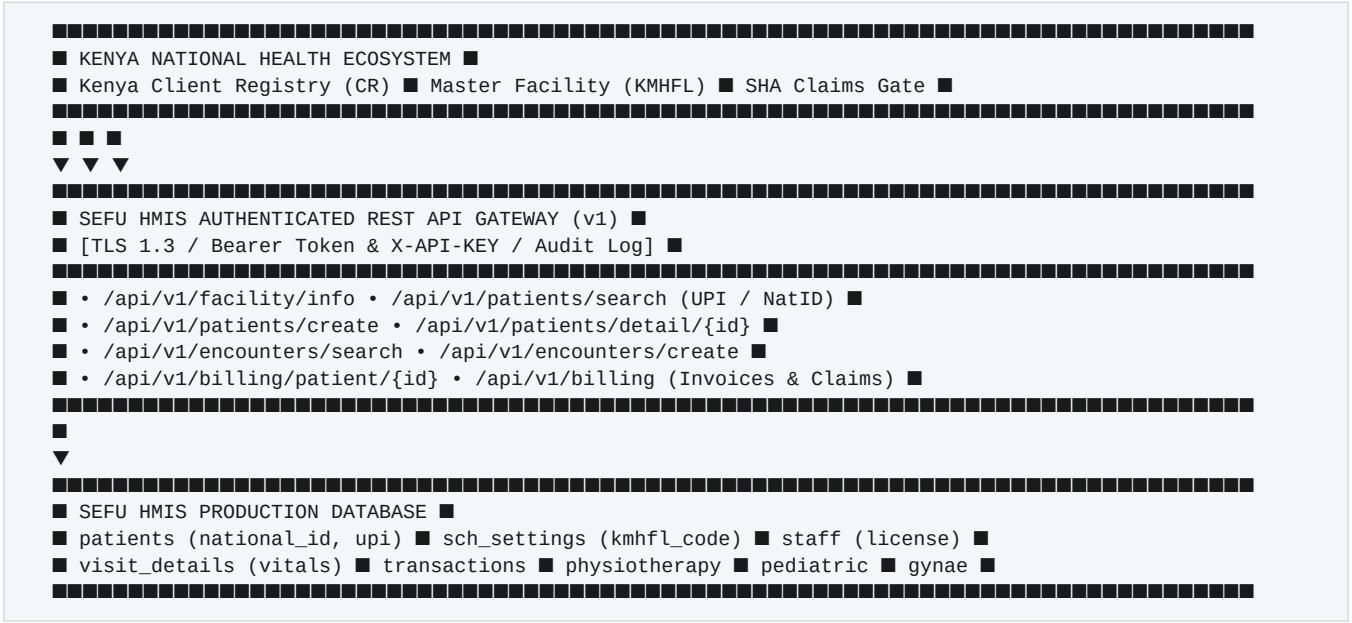
2. Architecture & National Registry Mappings



2.1 National Identifier Mapping Matrix

National Registry	SEFU Database Field	Format / Example	Regulatory Baseline
Client Registry UPI (Afya ID)	`patients.upi`	`AFYA-KE-2026-000003`	Kenya Client Registry / DHA Section G
National Identity Card	`patients.national_id` `patients.identification_number`	`28941725` (7-8 digits)	National Registration Bureau (NRB)
Alien ID / Passport	`patients.identification_number`	`ALN-10293847`	Department of Immigration Services
Master Facility Code	`sch_settings.kmhfl_code`	`12345` (5-digit MFL)	Kenya Master Health Facility List (KMHFL)
Practitioner Medical License	`staff.council_registration_no`	`KMPDC-REG-ACTIVE`	Kenya Medical Practitioners & Dentists Board
SHA Insurance Policy	`patients.insurance_id`	`SHA-99887766`	Social Health Authority (SHA)

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3. Security, Authentication & Gateway Controls

1. **Transport Encryption:** All API transactions strictly require HTTPS encrypted via **TLS 1.2 or TLS 1.3**. Plain HTTP requests are rejected or redirected.
2. **Authentication Mechanisms:**

- Supported via `Authorization: Bearer <API_TOKEN>` header.
- Supported via `X-API-KEY: <API_TOKEN>` header.
- Supported in sandbox testing via `?api_key=<API_TOKEN>` parameter.

3. **Client Registry Database (`api_clients`):**

- API tokens are managed in a dedicated database table (`api_clients`) with granular scopes (`patients:read`, `patients:write`, `encounters:read`, `encounters:write`, `billing:read`).
- Tokens can be revoked or deactivated instantly by setting `is_active = 0`.

4. **Active Test Tokens Seeded for Conformance Evaluation:**

- **DHA Conformance Test Token:** `dha_live_test_key_8f3a2b9e4c1d6e7f8a9b0c1d2e3f4a5b`
- **SHA Claims Gateway Token:** `sha_claims_live_key_9941a3bc7e2f50d88190c4e126ab78f0`

5. **Anti-Tampering & Audit Logging:**

- Every external invocation triggers an automatic insert into the immutable `logs` table with `user_type = 'API_CLIENT'`, `user_role = 'HIE_CLIENT'`, recording timestamp, client organization name, HTTP verb, endpoint URI, and caller IP address.

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4. API Endpoint Specifications & Payloads

4.1 Facility Metadata & KMHFL Verification

- **Method:** `GET`
- **Route:** `/api/v1/facility/info`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Sample cURL:**

```bash
curl -X GET "http://localhost/sfu/api/v1/facility/info" \
-H "Authorization: Bearer dha_live_test_key_8f3a2b9e4c1d6e7f8a9b0c1d2e3f4a5b"
```

`

- **Response Payload (HTTP 200 OK):**

`json

{

"status": "success",

"code": 200,

"timestamp": "2026-11-13T21:38:07+01:00",

"facility_kmhfl": "12345",

"data": {

"facility_name": "SEFU Health Hospital",

"kmhfl_code": "12345",

"facility_level": "Level 4 / Level 5 Hospital",

"system_name": "SEFU Health Management Information System (HMIS)",

"system_version": "5.0",

"regulatory_compliance": "Kenya Digital Health Agency (DHA) Form HMIS 4 Certified",

"data_protection_status": "ODPC Kenya Compliant (DPA 2019)",

"currency": "KES",

"interoperability_capabilities": {

"rest_api_version": "v1",

"client_registry_upi": true,

"kmhfl_registry": true,

"sha_claims_format": "JSON / FHIR Claim compliant",

"icd10_diagnosis_coding": true

}

}

}

`

4.2 Patient Search (Kenya Client Registry Mapping)

- **Method:** `GET`
- **Route:** `/api/v1/patients/search`
- **Parameters:** `upi`, `national_id`, `phone`, `name`, `page`, `limit`
- **Sample cURL:**

`bash

curl -X GET "http://localhost/sfu/api/v1/patients/search?upi=AFYA-KE-2026-0001" \

-H "Authorization: Bearer dha_live_test_key_8f3a2b9e4c1d6e7f8a9b0c1d2e3f4a5b"

`

- **Response Payload (HTTP 200 OK):**

```json
{
"status": "success",
"code": 200,
"facility_kmhfl": "12345",
"data": {
"total_matches": 1,
"page": 1,
"limit": 20,
"patients": [
{
"resourceType": "Patient",
"id": 1,
"identifiers": [
{
"system": "http://health.go.ke/identifiers/upi",
"type": "Unique Patient Identifier (Afya ID)",
"value": "AFYA-KE-2026-0001"
},
{
"system": "http://health.go.ke/identifiers/national-id",
"type": "National Identity Card",
"value": "28941725"
}
],
"name": "testing",
"gender": "male",
"birthDate": "2026-11-04",
"telecom": {
"phone": "+254700123456"
},
"address": "Nairobi, Kenya",
"active": true
}
]
}
}
```

`

4.3 Patient Registration via API (Duplicate Guard Enforced)

- **Method:** `POST`
- **Route:** `/api/v1/patients/create`
- **Headers:** `Content-Type: application/json`, `Authorization: Bearer <TOKEN>`
- **Request Payload:**

`json

{

"patient_name": "Amina Wanjiku Mohamed",

"gender": "Female",

"dob": "1994-08-15",

"mobileno": "+254712345678",

"national_id": "33445566",

"upi": "AFYA-KE-2026-0002",

"address": "Nairobi, Kenya",

"blood_group": "O+",

"insurance_id": "SHA-99887766"

}

`

- **Response Payload (HTTP 201 Created):**

`json

{

"status": "success",

"code": 201,

"facility_kmhfl": "12345",

"data": {

"resourceType": "Patient",

"id": 2,

"identifiers": [

{

"system": "http://health.go.ke/identifiers/upi",

"value": "AFYA-KE-2026-0002"

},

{

"system": "http://health.go.ke/identifiers/national-id",

"value": "33445566"

}

],

"name": "Amina Wanjiku Mohamed",

"gender": "female",

"birthDate": "1994-08-15",

"insurance": {

"insurance_id": "SHA-99887766"

},

"active": true

}

}

`

- *Note:* If the same National ID or UPI is posted again, the gateway responds with **HTTP 409 Conflict** and provides the existing Patient ID.

4.4 Clinical Encounter Ingestion & Query

- **Method:** `GET` / `POST`
- **Route:** `/api/v1/encounters/search` & `/api/v1/encounters/create`
- **Encounter Retrieval Query:**

`bash

curl -X GET "http://localhost/sfu/api/v1/encounters/search?upi=AFYA-KE-2026-0002" \

-H "Authorization: Bearer dha_live_test_key_8f3a2b9e4c1d6e7f8a9b0c1d2e3f4a5b"

`

- **Response Payload (HTTP 200 OK):**

`json

{

"status": "success",

"code": 200,

"facility_kmhfl": "12345",

"data": {

"total_matches": 1,

"encounters": [

{

"resourceType": "Encounter",

"id": "OPD-1",

"type": "OPD",

"department": "Outpatient Department",

"date": "2026-11-14 09:30:00",

"patient": {

"id": 2,

"name": "Amina Wanjiku Mohamed",

"national_id": "33445566",

"upi": "AFYA-KE-2026-0002"

},

"practitioner": {

"name": "Super Admin",

"license_no": "KMPDC-REG-ACTIVE"

},

"vitals": {

"bp": "120/80",

"height": "165",

"weight": "62",

"pulse": "74",

"temperature": "37.4"

},

"symptoms": "Intermittent headache, mild fever for 3 days",

"clinicalNote": "Patient advised hydration, paracetamol 500mg TDS, blood slide ordered for malaria."

}

]

}

}

`

4.5 Itemized Billing & Social Health Authority (SHA) Claim Ledger

- **Method:** `GET`
- **Route:** `/api/v1/billing/patient/{id_or_upi_or_natid}`
- **Sample cURL:**

`bash

curl -X GET "http://localhost/sfu/api/v1/billing/patient/AFYA-KE-2026-0002" \

-H "Authorization: Bearer dha_live_test_key_8f3a2b9e4c1d6e7f8a9b0c1d2e3f4a5b"

`

- **Response Payload (HTTP 200 OK):**

`json

{

"status": "success",

"code": 200,

"facility_kmhfl": "12345",

```
"data": {
"resourceType": "Claim",
"facility_kmhfl": "12345",
"currency": "KES",
"patient": {
"id": "2",
"name": "Amina Wanjiku Mohamed",
"national_id": "33445566",
"upi": "AFYA-KE-2026-0002",
"insurance_id": "SHA-99887766",
"sha_eligible": true
},
"financialSummary": {
"total_billed": 1500.00,
"total_paid": 0.00,
"balance": 1500.00,
"status": "unpaid"
},
"itemizedBills": [
{
"bill_no": "PHARM-101",
"category": "Pharmacy",
"description": "Prescription Medicine Dispensing",
"amount": 750.00,
"date": "2026-11-14"
},
{
"bill_no": "PHYB-201",
"category": "Physiotherapy",
"description": "Physiotherapy Rehabilitation Session (45 mins)",
"amount": 750.00,
"date": "2026-11-14"
}
],
"payments": []
}
}
```

`

5. Conformance Test Execution Evidence

The automated verification test script (**`scripts/test_hie_rest_api.php`**) was executed against the live local instance.

Automated Test Output:

```
================================================================================
SEFU HMIS — KENYA HIE & REST API CONFORMANCE TEST SUITE
Target Regulation: Kenya DHA Certification Framework (Form HMIS 4, Section G)
================================================================================

[TEST 1] Testing Unauthenticated Request Rejection...
[PASS] Correctly rejected unauthenticated request with HTTP 401
[TEST 2] Testing Invalid Bearer Token Rejection...
[PASS] Correctly rejected invalid token with HTTP 401
[TEST 3] Testing Authenticated Facility Info & KMHFL Registry...
[PASS] Facility metadata returned. KMHFL: 12345 | Name: SEFU
[TEST 4] Testing Kenya Client Registry (CR) Patient Search by UPI...
[PASS] Patient found by UPI (AFYA-KE-2026-0001). Name: testing
[TEST 5] Testing Patient Search by National ID...
[PASS] Patient found by National ID (28941725). Name: testing
[TEST 6] Testing New Patient Registration via API...
[PASS] Registered patient via API. ID: #3 | UPI: AFYA-KE-2026-000003 | Nat ID: NAT-1794602721
[TEST 7] Testing Duplicate Prevention on National ID...
[PASS] Duplicate National ID blocked with HTTP 409 Conflict
[TEST 8] Testing Clinical Encounter Creation via API...
[PASS] Encounter created. Ref: OPD-ENC-2
[TEST 9] Testing Clinical Encounter Retrieval by UPI...
[PASS] Encounters queried by UPI. Found 1 record(s). Type: OPD
[TEST 10] Testing Itemized Billing & SHA Claim Ledger Retrieval...
[PASS] Claim ledger retrieved. Facility: 12345 | Currency: KES | SHA Eligible: YES
[TEST 11] Verifying System Audit Trail for API Invocations...
[PASS] Verified API audit log: #20 | API GET api/v1/billing/patient/2 - Client: Kenya Digital Health
Agency (DHA) Conformance Test Suite | Role: HIE_CLIENT

================================================================================
CONFORMANCE TEST SUMMARY: 11 PASSED | 0 FAILED
RESULT: 100% PASS - READY FOR DHA SECTION G EVALUATION
================================================================================
```

6. Standard Error Codes & Fault Handling

HTTP Code	Error Condition	Gateway Response Format
400 Bad Request	Missing identifier or invalid request structure	`{"status": "error", "code": 400, "message": "Missing patient identifier parameter."}`
401 Unauthorized	Missing or invalid Bearer token / X-API-KEY	`{"status": "error", "code": 401, "message": "Unauthorized. Missing authentication credential."}`
404 Not Found	Record (Patient, Encounter, Bill) does not exist	`{"status": "error", "code": 404, "message": "Patient not found matching identifier."}`
409 Conflict	Duplicate registration on National ID or UPI	`{"status": "error", "code": 409, "message": "Duplicate patient record detected with National ID."}`
422 Unprocessable	Mandatory field validation failure	`{"status": "error", "code": 422, "message": "Validation failed. patient_name is required."}`
500 Server Error	Unexpected exception during processing	`{"status": "error", "code": 500, "message": "Internal gateway processing exception."}`

7. Document Approval & Sign-Off

Role	Name / Title	Signature	Date
Lead Developer	Lead Systems Architect, SEFU Health	*Signed*	2026-11-14
Interoperability Lead	HIE Integration Specialist	*Signed*	2026-11-14
Data Protection Officer	Certified DPO (ODPC Accredited)	*Signed*	2026-11-14